

Homework #1

PP 1.7

按关联参考方向, 电流从 "+" 流入, 计算功率 $P = U \cdot i$, 值为正表示消耗功率

① P_1 , $U = +5V$, $i = -9A \Rightarrow P_1 = -45W$

② P_2 , $U = +2V$, $i = 9A \Rightarrow P_2 = 18W$

③ P_3 , $U = +0.6I = +3V$, $i = 4A \Rightarrow P_3 = 12W$

④ P_4 , $U = +3V$, $i = 5A \Rightarrow P_4 = 15W$

验证: $P_1 + P_2 + P_3 + P_4 = -45 + 18 + 12 + 15 = 0W$ ✓

PP 2.2

根据欧姆定律

① $i = 3mA$, $R = 10k\Omega \Rightarrow U = i \cdot R = 30V$

② $G = 1/R = 0.1mS$

③ $P = U \cdot i = 30V \cdot 3mA = 90mW$

PP 2.6

根据 KVL 定律

$-70 + U_x + 2U_x - U_0 = 0 \dots ①$

根据欧姆定律 (记电流为顺时针 i)

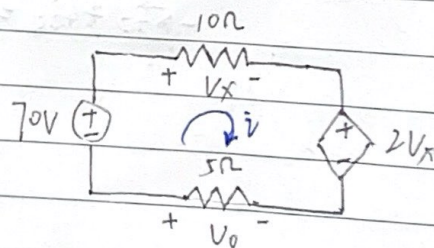
$U_x = 10i \dots ②$

$-U_0 = 5i \dots ③$

①
②
③ $\Rightarrow -70 + 30i + 5i = 0 \Rightarrow i = 2A$

$\Rightarrow U_x = 20V$

$U_0 = -10V$



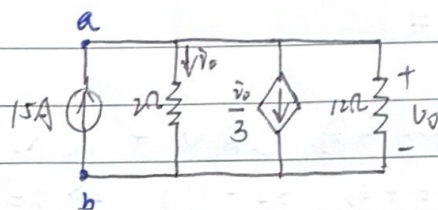
PP2-7

对节点 a 写 KCL 方程

$$15 = i_0 + \frac{1}{3}i_0 + \frac{V_0}{12} \quad \dots \textcircled{1}$$

对 2Ω 写欧姆定律方程

$$V_0 = 2 \cdot i_0 \quad \dots \textcircled{2}$$



$$\textcircled{1} \quad \Rightarrow 15 = \frac{4}{3}i_0 + \frac{1}{6}i_0 \Rightarrow i_0 = \frac{6}{9} \times 15 = 10 \text{ A}$$

$$\Rightarrow V_0 = 20 \text{ V}$$

PP2-8

对 a 点写 KCL:

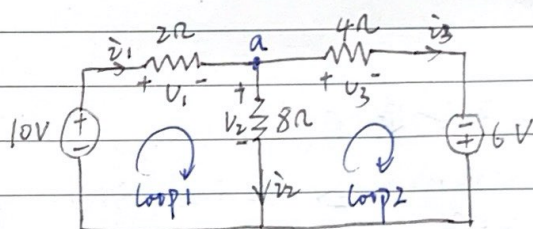
$$i_1 = i_2 + i_3 \quad \dots \textcircled{1}$$

对 loop1 写 KVL:

$$-10 + 2i_1 + 8i_2 = 0 \quad \dots \textcircled{2}$$

对 loop2 写 KVL:

$$-i_2 \cdot 8 + i_3 \cdot 4 - 6 = 0 \quad \dots \textcircled{3}$$



$$\begin{aligned} \textcircled{1} \quad & i_1 - i_2 - i_3 = 0 \\ \textcircled{2} \quad & 2i_1 + 8i_2 = 10 \\ \textcircled{3} \quad & -8i_2 + 4i_3 = 6 \end{aligned} \Rightarrow \begin{bmatrix} 1 & -1 & -1 \\ 2 & 8 & 0 \\ 0 & -8 & 4 \end{bmatrix} \cdot \begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 10 \\ 6 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix} = \begin{bmatrix} 3 \\ 0.5 \\ 2.5 \end{bmatrix} \text{ A}$$

$$\Rightarrow V_1 = 6 \text{ V}; V_2 = 4 \text{ V}; V_3 = 10 \text{ V}$$